Chapter 2: Life Processes in Living Organisms – Part 1

Q1. What are life processes? Name any four life processes. (3 Marks)
Subtopic: Living organisms and life processes
Life processes are the basic activities performed by living organisms to maintain life.
They are necessary for survival, growth and maintenance of the body.
Any four life processes are: nutrition, respiration, transportation, excretion, growth, reproduction.
Important points for full marks:
Write definition + list any four correctly.

Q2. Explain how living organisms produce energy. Write the significance of energy production. (4 Marks)
Subtopic: Living organism and energy production
Living organisms require energy to perform life processes such as movement, growth, repair and maintaining body functions.
Energy is produced mainly by breaking down food molecules inside cells during respiration.
The energy released is stored in the form of ATP, which is the energy currency of the cell.
Thus energy production is essential for continuous functioning and survival of organisms.
Important points for full marks:
Write ATP and energy currency of the cell.

Q3. Explain aerobic respiration with the help of a balanced chemical equation. (4 Marks)
Subtopic: Aerobic respiration
Aerobic respiration is the breakdown of glucose in the presence of oxygen to release energy.
It occurs mainly in mitochondria and produces carbon dioxide, water and a large amount of energy (ATP).
Balanced equation: $C_6H_{12}O_6 + 6O_2$ gives $6CO_2 + 6H_2O$ + Energy (ATP).
Aerobic respiration is efficient because complete oxidation of glucose occurs.
Important points for full marks:
Write the balanced equation correctly.
Mention oxygen, CO2, water and ATP.

Q4. Explain anaerobic respiration with suitable examples. (4 Marks)
Subtopic: Anaerobic respiration
Anaerobic respiration is the breakdown of glucose in the absence of oxygen to release energy.
It produces less energy because glucose is not completely broken down.
Example in yeast: glucose is converted into alcohol and carbon dioxide.
Example in human muscles during vigorous exercise: glucose is converted into lactic acid, causing tiredness and cramps.
Important points for full marks:
Write at least one example (yeast or muscles).
Mention less energy and incomplete breakdown.

Q5. Energy is obtained from different food components. Explain the role of carbohydrates, fats and proteins as energy sources. (5 Marks)
Subtopic: Energy from different food components
Carbohydrates are the primary and quick source of energy and are broken down into glucose for respiration.
Fats provide the maximum energy per unit mass and are used as long-term energy reserve; they are broken down into fatty acids and glycerol.
Proteins are mainly body-building foods used for growth and repair, but can also provide energy when carbohydrates and fats are insufficient.
Thus different food components contribute to energy production in different ways.
Balanced diet is important to provide all components in proper amount.
Important points for full marks:
Write that fats give maximum energy.
Write that proteins are mainly for growth and repair.

Q6. What is cell division? State its importance. (3 Marks)
Subtopic: Cell division
Cell division is the process by which a parent cell divides to form new daughter cells.
It is important for growth of an organism and increase in number of cells.
It helps in repair and replacement of damaged cells and in reproduction in some organisms.

Q7. Explain mitosis and write the main steps of mitosis. (5 Marks)
Subtopic: Mitosis and its steps
Mitosis is a type of cell division in which one parent cell divides to form two genetically identical daughter cells.
It occurs in somatic cells and is responsible for growth and repair.
Main steps of mitosis:
Prophase: chromosomes condense, nuclear membrane disappears and spindle fibres form.
Metaphase: chromosomes align at the equatorial plane.
Anaphase: sister chromatids separate and move to opposite poles.
Telophase: nuclear membranes re-form around each set of chromosomes and cytokinesis completes division.
Important points for full marks:

Write the four phases in order.
Mention two identical daughter cells.

Q8. What is meiosis? State its importance. (4 Marks)
Subtopic: Meiosis I and II
Meiosis is a special type of cell division in which one diploid cell divides to form four haploid cells.
It occurs in reproductive organs during formation of gametes.
It reduces chromosome number by half, maintaining constant chromosome number after fertilisation.
It introduces variations, which are important for evolution.
Important points for full marks:
Write four haploid cells and reduction division.
Mention variation.

Q9. Explain meiosis I and meiosis II briefly. (5 Marks)
Subtopic: Meiosis I and II
Meiosis occurs in two successive divisions: meiosis I and meiosis II.
Meiosis I is reductional division: homologous chromosomes separate, resulting in two haploid cells.
Crossing over and recombination occur in prophase I, producing variation.
Meiosis II is equational division: sister chromatids separate, similar to mitosis.
Finally four haploid daughter cells are formed.
Important points for full marks:
Write reductional (meiosis I) and equational (meiosis II).
Mention crossing over in prophase I.

Q10. Distinguish between mitosis and meiosis (any four points). (4 Marks)
Subtopic: Cell division; mitosis and meiosis
Mitosis occurs in somatic cells; meiosis occurs in reproductive cells.
Mitosis produces two identical diploid cells; meiosis produces four haploid cells.
Mitosis does not produce variation; meiosis produces variation due to crossing over.
Mitosis is for growth and repair; meiosis is for gamete formation and maintaining chromosome number.